

FuseLID: Vision-Language Model-Based Camera-Radar Fusion for Liquid Identification

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Abstract

Service robots often engage in liquid-related tasks such as pouring, which require accurate liquid identification. Vision-language models (VLM) exhibit strong performance in general object recognition, they struggle to reliably distinguish between visually similar liquids, limiting their effectiveness in such scenarios. As a promising solution, millimeter-wave (mmWave) radar combined with neural networks has been widely adopted for material recognition and liquid identification. However, training a radar-based liquid classifier demands a large volume of labeled radar-liquid data pairs and often faces challenges in generalizing to real-world environments. To address this challenge, we propose FuseLID, a VLM-based camera-radar late-fusion system designed to improve liquid identification performance. FuseLID leverages the weak penetration capability of mmWave signals through liquids and combines this with the robot's precise motion control to capture distinctive radar signatures of various liquids. Subsequently, a VLM-based late-fusion module is designed to combine camera and radar outputs for achieving enhanced liquid identification with limited radar-liquid data. Preliminary experiments show that FuseLID improves liquid identification accuracy from 51.6% to 95.7% when classifying several commonly consumed beverages, including visually similar ones.

CCS Concepts

• Human-centered computing → Ubiquitous and mobile computing systems and tools.

Keywords

Liquid identification, vision language model, mmWave radar, camera-radar fusion.

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Figure 1: VLMs consistently misclassify visually similar liquids such as Fanta (left) and orange juice (right), highlighting their limitations in fine-grained liquid recognition.

1 Introduction

Service robots, an important category of robotics, frequently handle liquid-related tasks like pouring or refilling a cup of drink, which often require accurate liquid identification. Vision-language models (VLM) exhibit strong capabilities to recognize objects across diverse workspaces, but they exhibit fundamental limitations in distinguishing visually similar liquids (e.g., color, transparency). Our preliminary test (Figure 1), along with findings from prior studies [3, 5, 6], demonstrates the limitations of VLMs in these scenarios, primarily due to the absence of distinctive optical features.

To address this limitation in optical perception, researchers have turned to millimeter-wave (mmWave) radar sensing as an alternative modality. Radar signals are sensitive to a material's dielectric properties, which vary significantly across common materials [2]. Distinct from lower frequency signals, mmWave signals have shorter wavelengths, resulting in limited penetration and stronger surface scattering when interacting with liquids. These properties highlight differences in radar reflection signatures across various liquid types [8]. Consequently, liquid identification using mmWave radar has seen significant advancements, producing promising results [8, 10, 13]. However, existing methods depend on extensive radar-liquid data pairs for model training, which might hinder their real-world deployment. In this context, we pose the following question: *Can the robust semantic understanding capabilities of VLMs be effectively combined with the material sensitivity of mmWave radar to enable fine-grained liquid identification?*

While conceptually feasible, developing a VLM and radar fusion system faces several challenges: (1) Heterogeneous Fusion: The fundamentally distinct characteristics of VLM-extracted image features and mmWave radar data necessitate robust fusion strategies. Additionally, VLM cannot directly process raw radar data. (2) Scarce Paired Datasets: To date, no publicly available large-scale dataset exists that pairs RGB images with mmWave radar data for the task

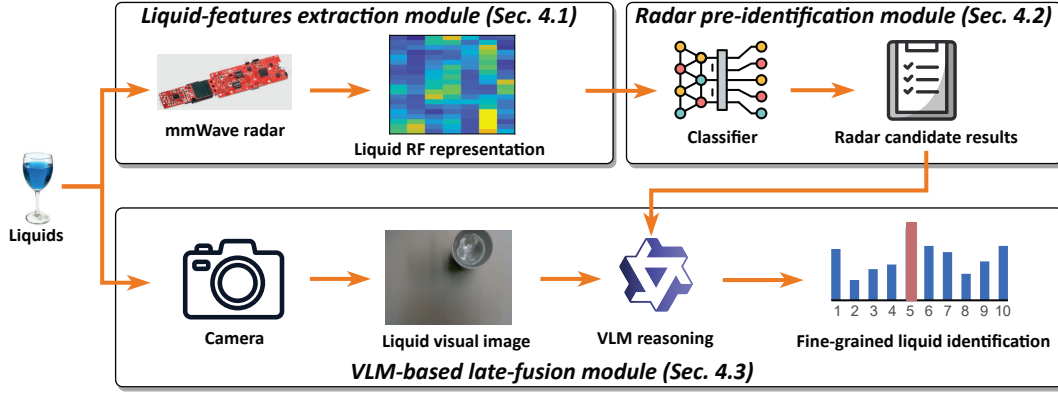


Figure 2: The architecture of FuseLID.

of liquid identification. Furthermore, sparse radar reflection points fail to fully capture liquid characteristics.

To overcome these challenges, we present FuseLID, the first VLM-based camera-radar late-fusion system designed to enable enhanced liquid identification for service robots, as illustrated in Figure 2. FuseLID exploits the robot’s precise motion control to synthesize a large antenna aperture, significantly enhancing radar signal resolution and information richness, helping overcome data sparsity. To address the challenges of heterogeneous sensor fusion and limited paired data, FuseLID adopts a late-fusion architecture. Specifically, it first trains a radar classification network independently using only our self-built radar-liquid dataset, eliminating the need for paired visual inputs during the radar model’s training phase. Subsequently, FuseLID leverages the strong reasoning capabilities to analyze candidate results from visual and radar-based classifiers, improving liquid classification accuracy by exploiting the complementary strengths of multi-modal data. This design enables robust identification performance, even when trained on a limited radar–liquid dataset. Preliminary experimental validation on a real-world robotic platform demonstrates that FuseLID significantly outperforms both VLM-only and radar-only baselines in distinguishing visually or chemically similar liquids.

2 Related Work

Accurate liquid identification, particularly in domestic environments, is essential for enabling context-aware, proactive interactions in service robots, allowing them to respond effectively to user needs and ongoing activities. Vision-based methods, empowered by large-scale VLMs, have led to diverse theoretical advancements and practical applications in liquid detection. However, these approaches often rely heavily on favorable visual conditions, such as adequate lighting, distinct liquid color, and transparency. As a result, they frequently struggle to distinguish between visually similar liquids. To address these limitations, radio frequency (RF) sensor-based techniques have been proposed, offering robust liquid recognition by capturing material-specific dielectric properties and other intrinsic physical parameters [4, 7, 9, 11–13]. Specifically, Tagscan utilizes received signal strength indicators (RSSI) and phase variations obtained through radio frequency identification (RFID) interrogations to build classification models for 10 distinct

liquid categories [12]. FG-LiquidID designs a novel neural network for liquid sensing using millimeter-wave radar, enabling fine-grained identification of 30 different liquid types [7].

Different from existing approaches, our proposed FuseLID employs a VLM-based late-fusion architecture that synergistically integrates camera-based and radar-based outputs for enhanced liquid recognition accuracy. FuseLID is designed to use VLM-derived visual cues to refine predictions from dedicated radar classifiers through semantic guidance. By exploiting the pretrained knowledge embedded in VLM, our approach demonstrates robust performance even with limited radar samples, particularly excelling in distinguishing visually similar liquids.

3 Motivation of The Late Fusion

Multi-modal fusion strategies are typically classified as early-fusion, middle-fusion, or late-fusion. We adopt late-fusion primarily due to the following two reasons: 1) Data dimension discrepancies. Typical visual data (640×480) and our radar representation (7×20) exhibit significant dimensional mismatches. This scale disparity makes network design for comprehensive feature extraction in early/middle-fusion architectures technically impractical. 2) Dataset size limitations. Although directly training a radar model for liquid identification is a straightforward solution, it requires more than ten thousand radar-liquid data pairs [15]. The late-fusion approach effectively leverages the prior knowledge embedded in VLMs to refine the outputs of the radar model, enabling performance comparable to large-scale dataset training, while requiring only limited radar data.

4 System Design

FuseLID consists of three major modules, as illustrated in Figure 2. First, a liquid-features extraction module employs a beamforming algorithm to obtain the RF representation of various liquids. Second, a neural network-based radar pre-identification module is employed to identify candidate liquids from their RF signatures, trained on a small-scale, custom-built radar-liquid dataset. Finally, the VLM-based late-fusion module integrates visual and radar modalities during inference by leveraging the strong reasoning capabilities of the pre-trained VLM to selectively filter and refine outputs from

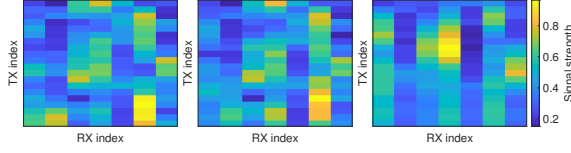


Figure 3: RF representation of three liquids. Fanta (left) and Sprite (middle) appear similar, whereas water (right) is clearly distinct.

the radar classifier, thereby enhancing the accuracy of liquid identification.

4.1 Liquid-Features Extraction Module

As noted in [14], the orientation of the radar antenna significantly affects the quality of material feature extraction. To address this, we mounted the mmWave radar on the robot's manipulator, enabling it to automatically position the sensor above the containers and orient it toward the liquid for optimal signal acquisition. Since mmWave signals result in minimal penetration and strong surface scattering within liquids, the dominant path of the received radar signal encapsulates the liquid's RF features. Consequently, in this module, we employ a beamforming algorithm to identify the dominant path and extract the corresponding reflected signal as the RF representation of the liquid.

4.1.1 Beamforming Algorithm. Let θ_l and τ_l denote the angle of arrival (AoA) and time of flight (ToF) of the l^{th} path, respectively. We assume the number of receiver antennas is M , arranged in a uniform linear array, and the radio signal is transmitted over K subcarriers. The phase shift vector can be expressed as

$$a(\theta_l, \tau_l) = [1, \dots, \phi_{mk}(\theta_l, \tau_l), \dots, \phi_{MK}(\theta_l, \tau_l)]^T, \quad (1)$$

where $\phi_{mk}(\theta_l, \tau_l)$ represents the phase shift at the k^{th} subcarrier sample and the m^{th} antenna of l^{th} path [1, 16].

Assuming L paths exist in the environment, the steering matrix is defined as $A = [a(\theta_1, \tau_1), a(\theta_2, \tau_2), \dots, a(\theta_L, \tau_L)]$. The received signal matrix is given by $H(t) = As(t) + n(t)$, where $s(t)$ is the emitted mmWave signal and $n(t)$ denotes background noise.

To extract the reflected paths, we discretize the AoA-ToF space into a grid, defining the corresponding steering matrix as

$$\hat{A} = [a(\hat{\theta}_1, \hat{\tau}_1), a(\hat{\theta}_2, \hat{\tau}_2), \dots, a(\hat{\theta}_N, \hat{\tau}_N)], \quad (2)$$

where N represents the number of AOA-ToF grid points.

Then, the AoA-ToF spectrum is computed by projecting $\hat{A}$ onto the received data $H(t)$, i.e.,

$$P = \hat{A}^H H(t) = \hat{A}^H A s(t) + \hat{n}(t), \quad (3)$$

where $A \in \mathbb{C}^{MK \times L}$, $\hat{A} \in \mathbb{C}^{MK \times N^2}$, $s \in \mathbb{C}^{L \times 1}$, and $\hat{n}(t) = \hat{A}^H n(t)$.

According to Eq. (3), with a dense AoA-ToF grid, i.e., $N \gg L$, the AoA-ToF spectrum P will have large amplitude on the grid points corresponding to the reflected paths. The AoA-ToF of the dominant path is the one with the maximum amplitude by excluding the line-of-sight (LOS) path, i.e.,

$$(\theta^*, \tau^*) = \arg \max_{(\hat{\theta}_l, \hat{\tau}_l) \neq (\theta_0, \tau_0)} |P| = |\hat{A}^H H(t)|, \quad (4)$$

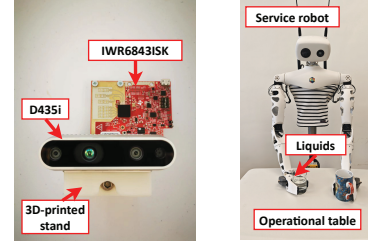


Figure 4: Hardware implementation (left) and the service robot workspace (right).

where (θ_0, τ_0) denotes the AoA-ToF of the LOS path. Then, the signal from the dominant path can be expressed as

$$\tilde{s}_d(t) = a(\theta^*, \tau^*)^H H(t). \quad (5)$$

4.1.2 RF Representation of Liquids. Recall the Friis transmission formula is given by

$$A_d = A_0 g_t g_r \frac{\lambda}{4\pi R} \Gamma_{liquid}, \quad (6)$$

where A_0 and A_d denote the amplitude of the transmitted and received signal at propagation distance R , respectively. Here, g_t and g_r represent the transmit and receive antenna gains, λ is the wavelength, and Γ_{liquid} is the reflection coefficient of the liquid.

With all other parameters fixed, the signal amplitude is directly proportional to the liquid's reflection coefficient. Therefore, the RF representation for any liquid can be derived using $\tilde{s}_d(t)$ and (θ^*, τ^*) obtained by Eq. (4) and (5). Meanwhile, by leveraging the robot's precise motion to synthesize a large antenna aperture, we can obtain a 7×20 Tx-Rx pairs channel matrix that enables more accurate representation and differentiation of liquids. Figure 3 illustrates three liquids' RF representation.

4.2 Radar Pre-Identification Module

By leveraging the liquid-feature extraction module, we obtain the RF representation corresponding to different liquid types. In this subsection, we propose a U-Net-based liquid pre-identification module that adopts convolutional neural networks (CNNs) and skip connections to classify liquids from their RF representations.

4.2.1 Network Architecture. The radar encoder employs a hierarchical CNN architecture to extract high-level latent representations from 2D radar inputs. The encoder is composed of stacked basic blocks, each comprising a 2D convolutional layer, batch normalization, and a ReLU activation function. The number of channels increases progressively in deeper layers to capture richer and more diverse features. The decoder adopts a symmetrical design to the encoder, with a decreasing number of channels and increasing output dimensions across stacked blocks. Dropout is applied to mitigate overfitting during high-dimensional expansion. A linear layer outputs logits for four classes using softmax activation.

4.2.2 Loss Function Design. To train the network for liquid identification from radar feature matrices, we adopt cross-entropy as the loss function. Cross-entropy directly quantifies the discrepancy between the model's output probability distribution and the ground

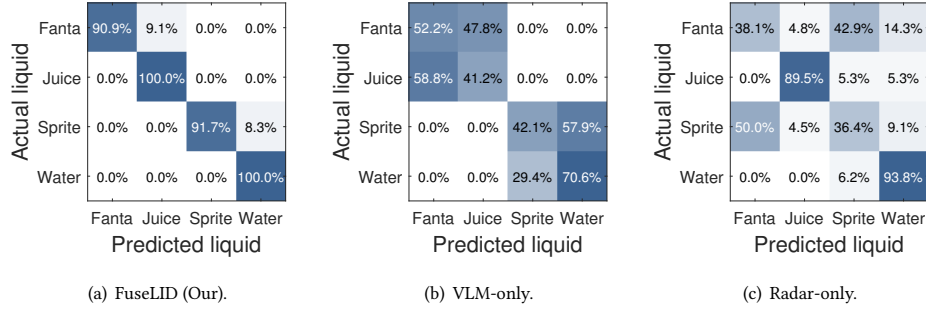


Figure 5: Confusion matrices of liquid identification.

truth labels. In multi-class classification tasks, it effectively guides model optimization and enhances training efficiency.

4.2.3 Self-built Radar-Liquid Dataset. We made a new dataset comprising RF representations of liquids. The dataset consists of 400 realistic radar-liquid data pairings, encompassing four commonly encountered liquids in daily life: Fanta, orange juice, Sprite, and water. Each sample includes one liquid category, with controlled variations in container types and liquid levels, to reflect the diversity of real-world environments commonly encountered by service robots. We collect the radar data while both the robot and the target are static. We remark that as we can regard the RF representation as a type of image, standard data augmentation techniques can also be applied to these mmWave samples.

4.3 VLM-Based Late-Fusion Module

As discussed in Section 1, while designing a comprehensive end-to-end network is a straightforward approach for liquid identification, it requires extensive labeled data for training. To address this limitation, FuseLID employs a VLM-based visual-radar late-fusion approach to enable robust liquid identification, while using limited radar data.

We exploit the pre-trained VLM’s ability to extract semantic information from visual data, which provides prior knowledge about liquid properties. Although VLM struggle to distinguish visually similar liquids, their ability to differentiate liquids with distinct visual characteristics enables refinement of the radar pre-identification module’s predictions, thereby significantly reducing the amount of training data required. The output of the pre-identification module was modified to generate both the most probable candidate predictions and their associated confidence scores. Subsequently, the late-fusion approach integrates candidate results from the radar classifier and visual data into the VLM, leveraging the VLM’s strong reasoning capabilities to analyze intermediate results and improve liquid identification accuracy.

To validate our approach, we conducted a preliminary experiment involving the identification of a cup of Fanta on a table. The radar classifier was trained on 200 samples (50 per category: Fanta, orange juice, Sprite, and water), followed by direct prediction using extracted radar features. Due to the limited dataset size, the network experienced severe overfitting during training, resulting in poor generalization during testing. Consequently, the classifier

classified the liquid as either Fanta or Sprite, a result consistent with expectations, as these weakly acidic liquids exhibit similar radar features. Meanwhile, the VLM identified the orange-colored liquid as either Fanta or orange juice via visual analysis, assigning higher confidence to Fanta. Through the late-fusion and VLM reasoning framework, the system ultimately classified the liquid as Fanta.

5 Experimental Evaluation

5.1 Experiment Setup

In this section, we evaluate FuseLID using different liquids. We implement FuseLID using a TI IWR6843ISK mmWave radar, which operating at a center frequency of 61.8 GHz and a bandwidth of 3.6 GHz. The mmWave radar unit is mounted alongside an Intel Realsense D435i RGB-D camera on a custom-designed 3D-printed stand, fixed on the hand of Reachy robot, as shown in Figure 4. With robot’s precise motion to synthesize a large antenna aperture, we can obtain a 7×20 Tx-Rx pairs. The virtual antenna array must face the liquid to avoid errors introduced by antenna position. The entire system operates on Ubuntu 20.04 and interfaces with the devices through Ethernet connections for robust data transmission and control.

5.2 Overall Performance

We now present the liquid identification performance of FuseLID, and further compare it with a VLM-only¹ and a radar-only identification algorithm. We prepared four liquids that appear in our self-built dataset: Fanta, orange juice, Sprite, and water. We fill the cup halfway for each liquid, as too little liquid will reduce the quality of the RF representation. Additionally, we employ two distinct paper cups to facilitate liquid identification from top-view radar images, ensuring that classification relies on the liquid’s properties rather than the container’s appearance.

We use confusion matrices in Figure 5 to show and compare the identification performance of different methods. We observe that the late-fusion solution used in FuseLID is the most robust compared with the VLM-only and radar-only methods. The overall accuracy for identifying the liquids is 95.7%, substantially higher than the VLM-only (51.6%) and radar-only (64.5%) methods. Notably, FuseLID prioritizes visual data during the VLM-based late-fusion

¹We use Qwen2.5-VL-7B model and visual input for liquid identification.

phase, regarding radar candidates as supplementary information. Despite these improvements, certain visually similar liquids remain challenging to distinguish, leading to occasional misclassifications. As shown in Figure 5(a), a few Fanta samples were misclassified as orange juice, and some Sprite samples were incorrectly identified as water.

The VLM-only method enables coarse-grained correct classification of liquids based on color and transparency, yet fails to achieve fine-grained liquid identification, particularly for visually similar liquids. Figure 5(b) presents the confusion matrix of the VLM-only method. Benefiting from the extensive pretraining data of VLM, it can be observed that the VLM-only method correctly identifies transparent liquids as Sprite or water, yet cannot distinguish between specific liquid types. Figure 5(c) demonstrates the confusion matrix of the radar-only approach. Here we employed 400 training samples, showing that radar-only methods easily distinguish juice from water, yet cannot differentiate Fanta and Sprite, which share similar acidic properties.

6 Limitations and Future Work

Category Limitations. The current work focuses on liquid recognition for only 4 categories, fewer than prior studies, and limited radar-liquid data pairs also constrain FuseLID's performance. In future work, we plan to collect more liquid categories and extend liquid identification to over 16 types, further enhancing practical applicability.

Optimizing the Late-Fusion Strategy. As demonstrated in Section 5.2 and Figure 5(a), the current fusion strategy prioritizes visual data, which can lead to identification errors when visual and radar-based identification results conflict. In future work, we aim to refine the late-fusion strategy by leveraging the VLM's reasoning capabilities through techniques such as low-rank adaptation (LoRA) fine-tuning. This will enable FuseLID to more effectively reason visual and radar outcomes to further improve the performance.

Model Overfitting. As shown in Figure 5(c), the average accuracy of the current radar liquid identification module is 64.5% on the test set, which is worse than the results on the training. Fewer training samples can lead to model overfitting. Therefore, we will collect more data to train the model. And standard data augmentation techniques can also be applied to these samples.

7 Conclusion

Reliable liquid identification is essential for enabling service robots to perform complex tasks effectively. While VLM demonstrates strong capabilities in object detection, their performance declines when distinguishing visually similar liquids. Although mmWave radar has shown potential for liquid sensing, its practical deployment is hindered by data-intensive training. Our FuseLID framework addresses these limitations by enabling robust extraction of RF-based liquid representations from radar data and leveraging the reasoning capabilities of VLMs to optimize cross-modal integration, thereby achieving fine-grained liquid identification. Preliminary experimental results demonstrate that FuseLID achieves a substantial accuracy improvement (from 51.6% to 95.7%) in identifying visually similar liquids, surpassing existing methodological constraints and offering a practical, efficient solution for precise liquid perception.

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